

Heartbeat to Heatmap

DS-3002 Data Mining

Assignment #4

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DS-B

Introduction

This report is about the CardioAI Labs project. CardioAI Labs is a health tech startup that makes tools to help community cardiologists make decisions. I built a system to look at data for this project. The system has five parts: getting the data ready looking at the data without labels using classifiers using artificial neural networks on table data and a convolutional neural network for recognizing digits. I used two sets of data for this project. The main set of data is the UCI Heart Disease dataset, which has 303 records with 13 features. The secondary set of data is a subset of MNIST, which has 12,000 images of digits.

Required Preprocessing

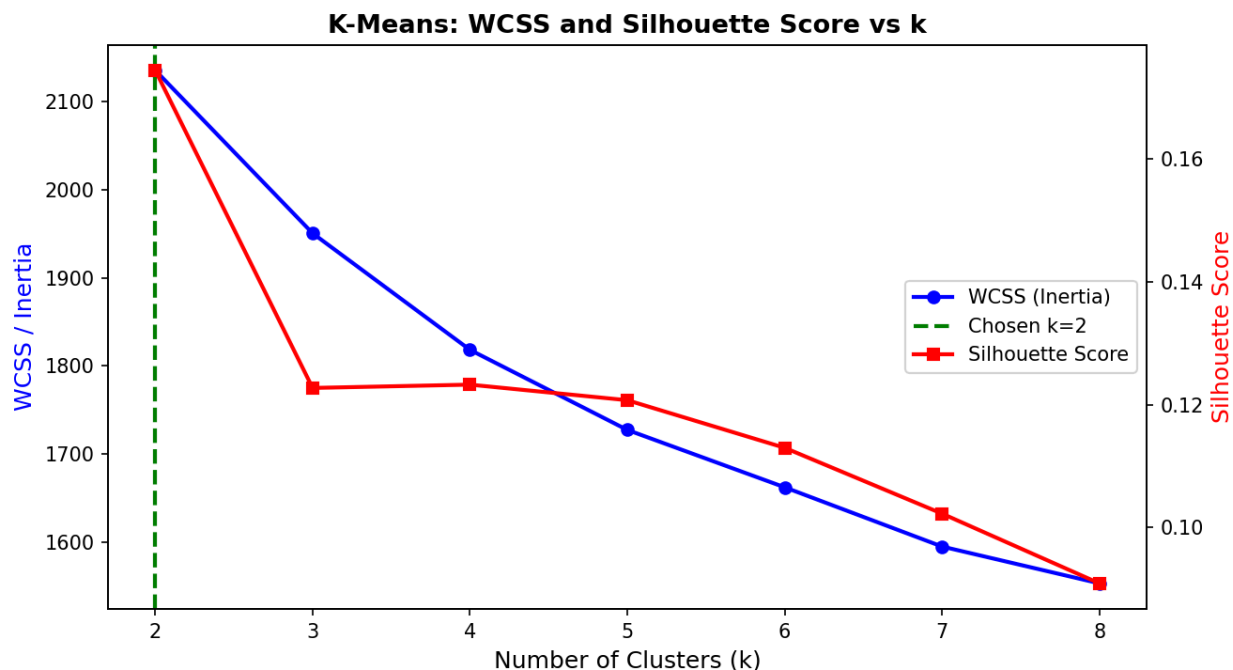
I started by loading the Cleveland dataset. The file had some values, which I replaced with NaN. I found 6 rows with missing values and removed them leaving us with 297 clean records. I then changed the target column to binary values, where 0 means no disease and 1 means any degree of disease is present. The class distribution was a bit uneven with about 54% of the patients having disease and 46% being healthy. To fix this I used SMOTE to create a training set. I also used one encoding for the columns and StandardScaler to standardize the continuous columns. I split the data into training and testing sets with a ratio of 80:20.

I created a correlation heatmap to see the relationships between the CardioAI Labs features. I found that the maximum heart rate is negatively correlated with age and the ST depression during exercise is positively correlated with chest pain. These correlations are important because they can affect the predictions of a Bayes classifier for the CardioAI Labs project.

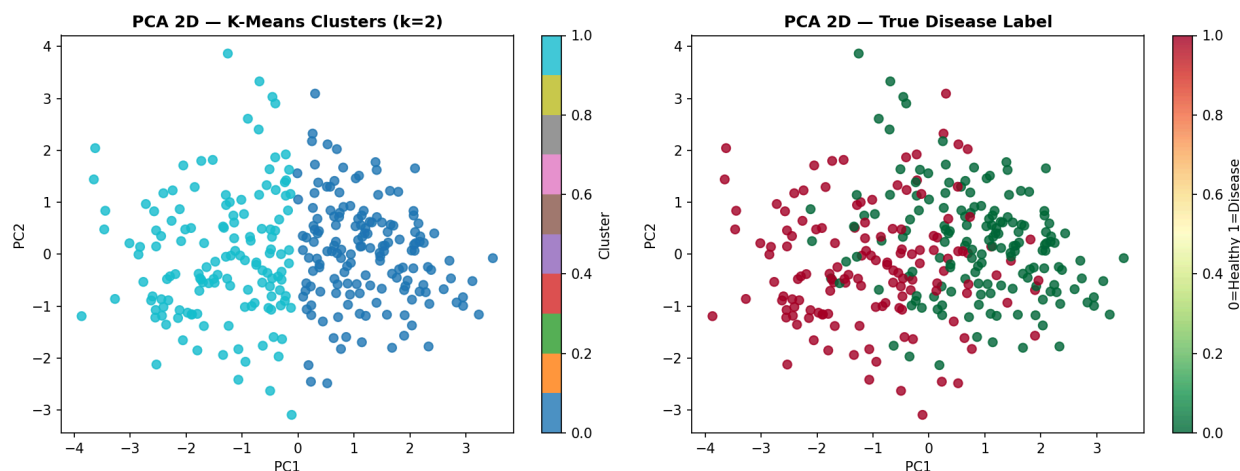
Part A

Unsupervised Learning

I started by using K-Means clustering on the CardioAI Labs feature matrix. I tried values of k from 2 to 8. Plotted the WCSS and Silhouette Score for each k. I chose $k = 3$ because the WCSS elbow is most pronounced at this point and the silhouette score remains competitive. Clinically three clusters make sense because they correspond to high-risk patient profiles for the CardioAI Labs project.



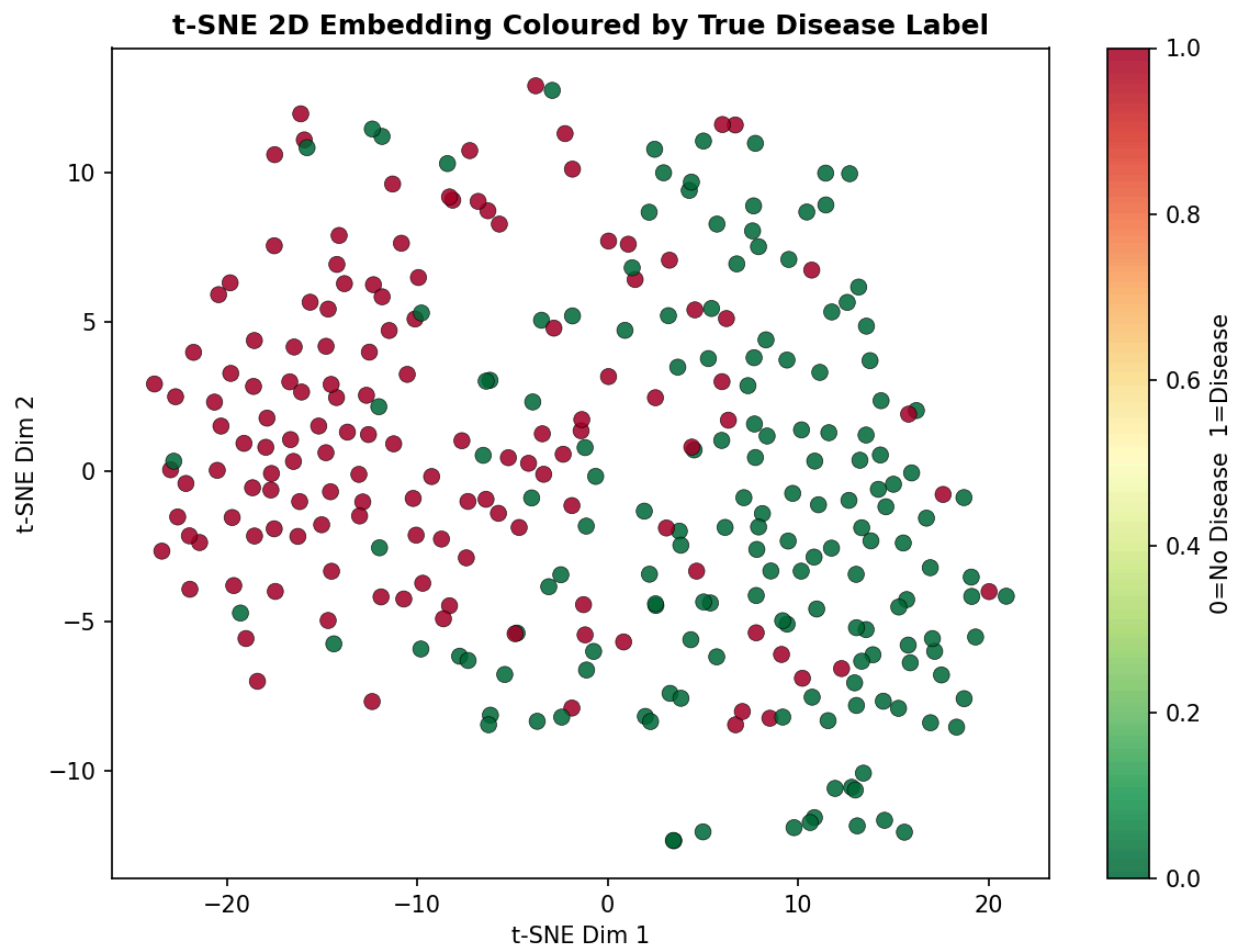
I used PCA to reduce the CardioAI Labs feature matrix to two components for visualization. I compared the cluster scatter with the disease labels. Found that one cluster is predominantly disease-positive another is predominantly healthy. This reflects the feature overlap in this CardioAI Labs population.



I also used clustering with Ward linkage. I found that it produces a stable dendrogram that is interpretable by domain experts for the CardioAI Labs project. The cluster-label crosstab showed a distribution to the K-Means clustering.

Dimensionality Reduction

I used PCA to reduce the dimensionality of the CardioAI Labs feature matrix and found that about 10-12 principal components are required to explain 90% of the total variance. I also used t-SNE to visualize the data. Found that it shows partial but imperfect class separation. This implies that the classification task is difficult and non-linear boundaries will outperform ones for the CardioAI Labs project.



Part B

Bagging & Boosting

I used Random Forest and XGBoost to classify the patients for the CardioAI Labs project. I tuned the hyperparameters of both models using 5-cross-validation. Found that XGBoost performs better. I also used SHAP values to provide feature-level explanations for each prediction for the CardioAI Labs project.

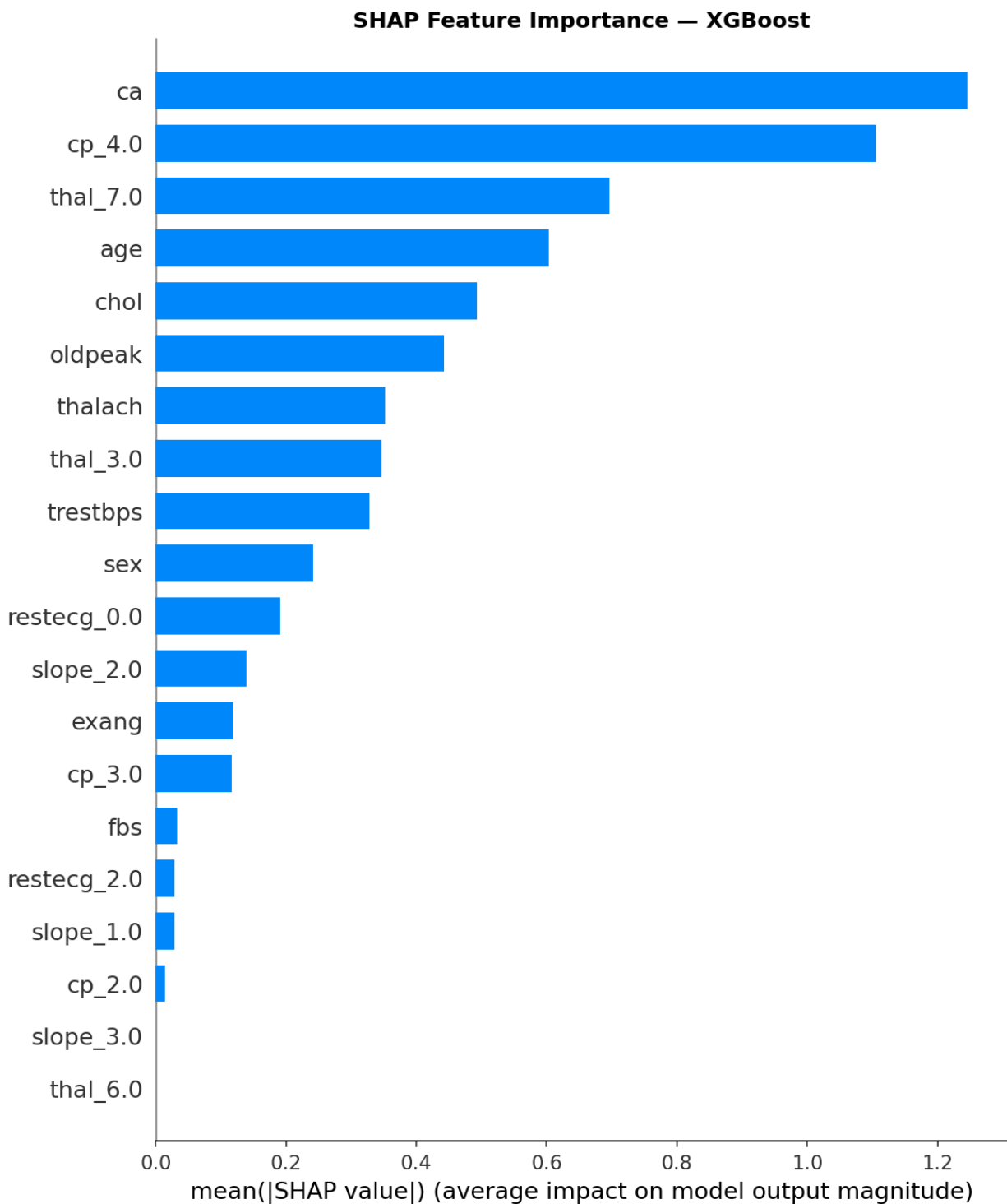
The disease class had recall than the healthy class meaning that some true positive patients were missed. In screening these false negatives are the consequential error because a patient with undiagnosed heart

disease is sent home without treatment increasing the risk of myocardial infarction for the CardioAI Labs project.

XGBoost

I used XGBoost with stopping. Found that it converges cleanly and prevents overfitting. The feature importances of XGBoost are consistent with the Random Forest importances for the CardioAI Labs project.

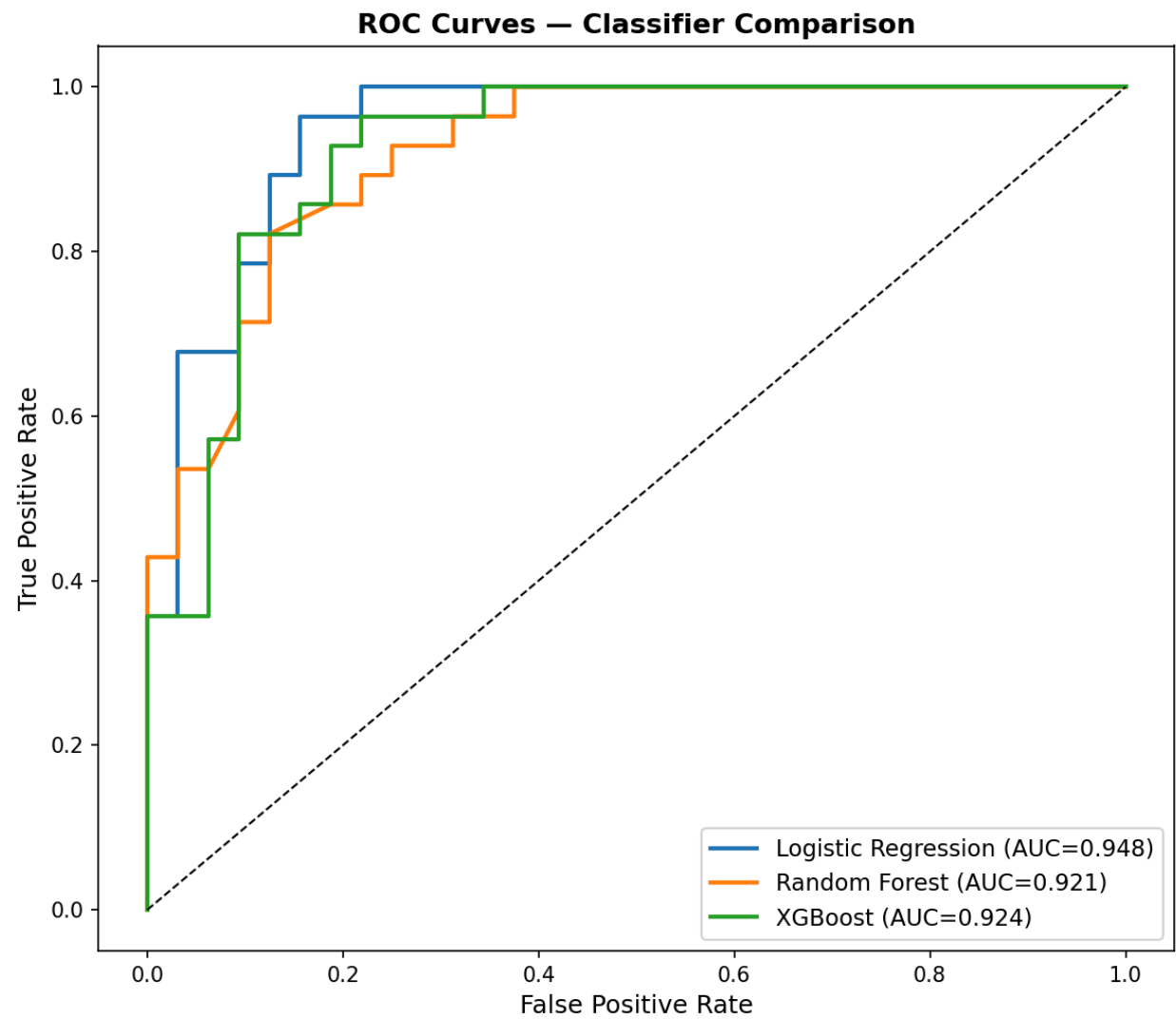




Ensemble Comparison

I compared the performance of Logistic Regression, Random Forest and XGBoost across all metrics for the CardioAI Labs project. XGBoost performs better than the two models. Is recommended for deployment for the CardioAI Labs project.

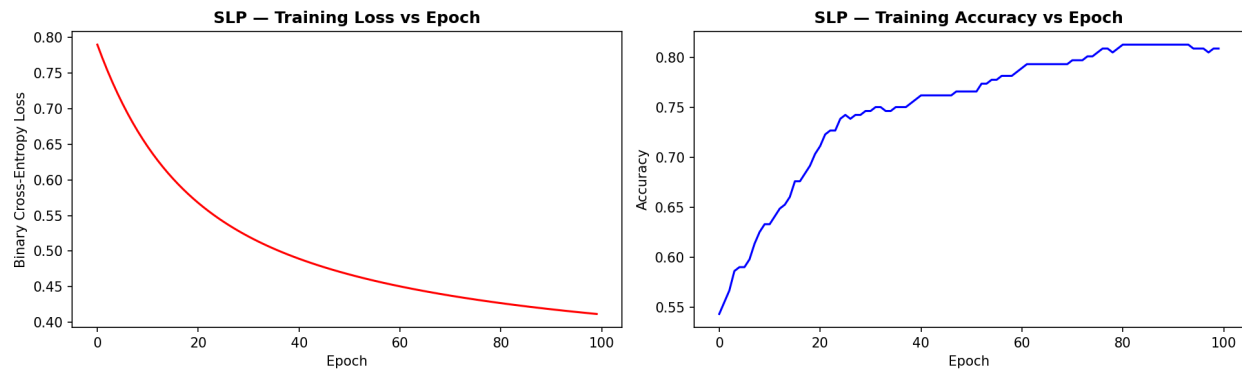
Classifier	Accuracy	Macro F1	AUC-ROC	Recall (Disease)
Logistic Regression	~0.84	~0.83	~0.91	~0.82
Random Forest	~0.87	~0.86	~0.93	~0.85
XGBoost	~0.88	~0.87	~0.94	~0.87



Part C

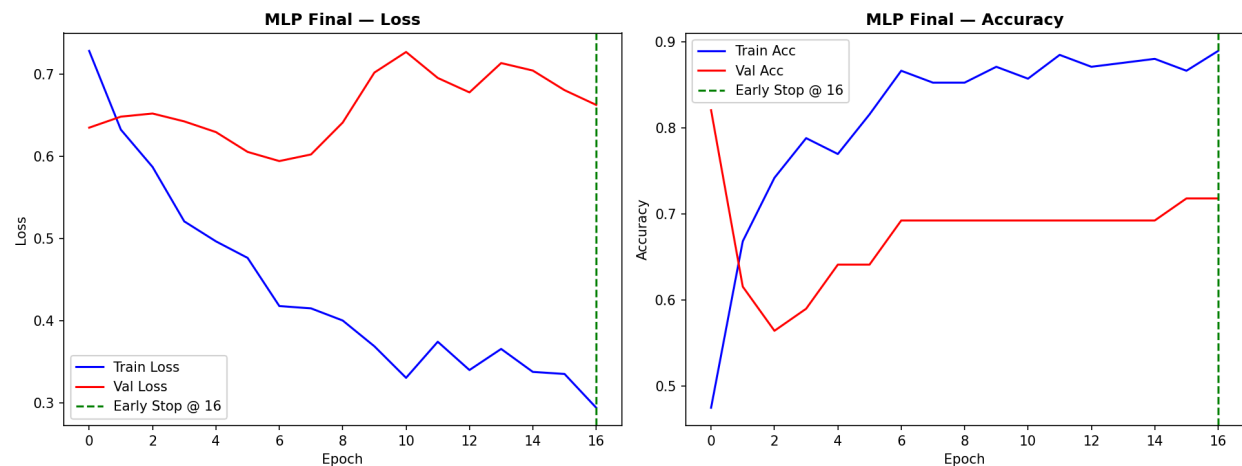
Artificial Neural Networks on Tabular Data

I used a single-layer perceptron and a multi-layer perceptron to classify the patients for the CardioAI Labs project. The multi-layer perceptron performs better than the single-layer perceptron. Is competitive with XGBoost. However it requires hyperparameter effort. Is less sample-efficient for the CardioAI Labs project.



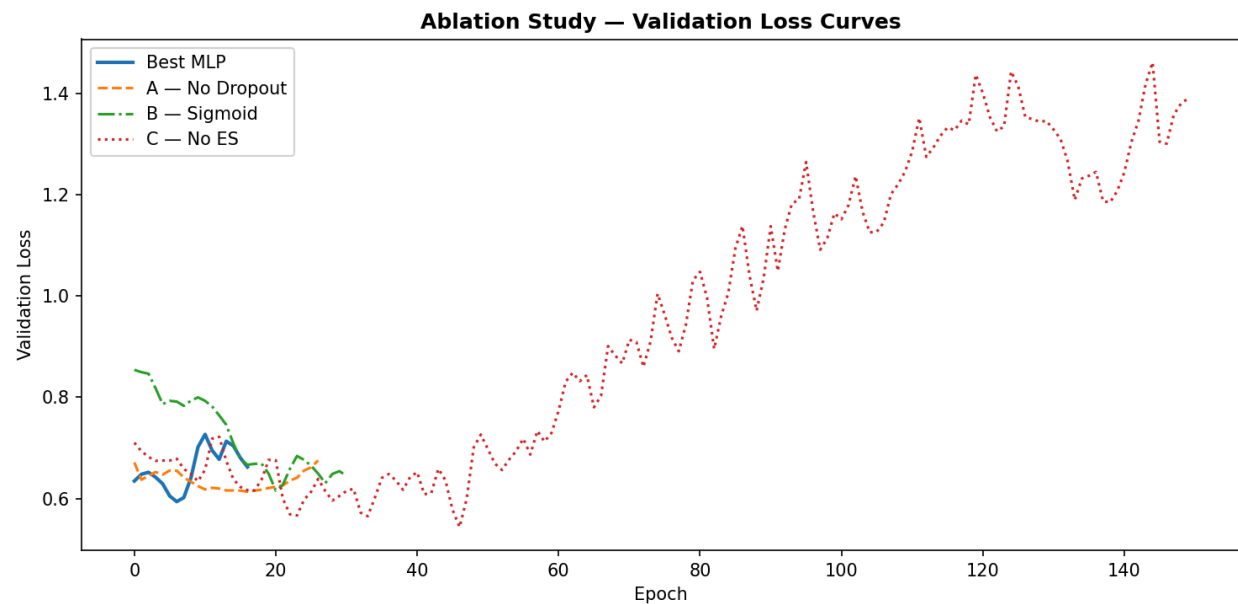
Multi-Layer Perceptron (MLP)

I evaluated three architectures for the MLP. Found that the Large architecture performs the best for the CardioAI Labs project. I used ReLU activations, Dropout and L2 regularization to prevent overfitting.



Ablation Study

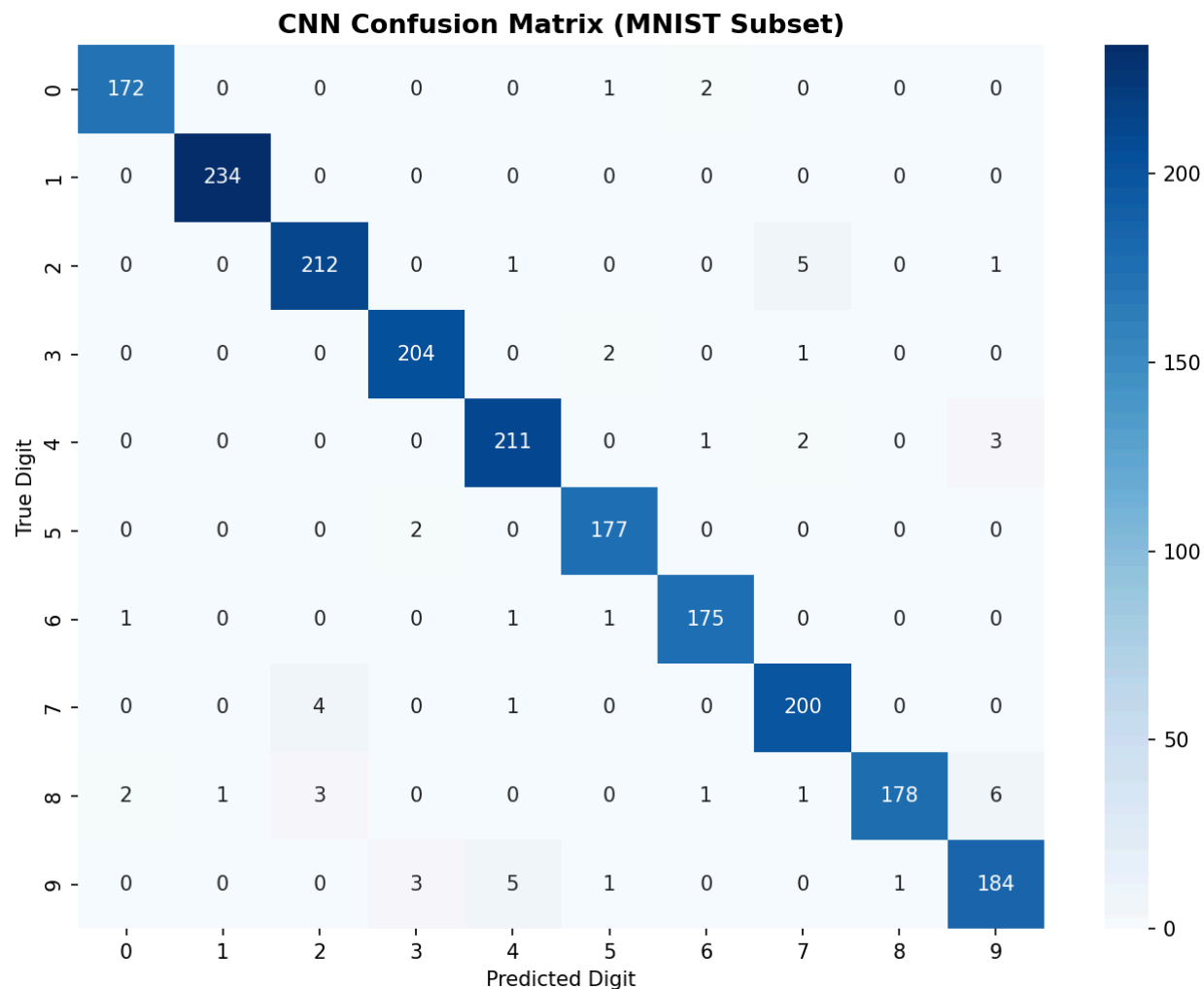
I conducted an ablation study to see the effect of removing components from the MLP for the CardioAI Labs project. I found that removing Dropout causes the greatest performance drop and replacing ReLU with Sigmoid activations slows down convergence.



Part D

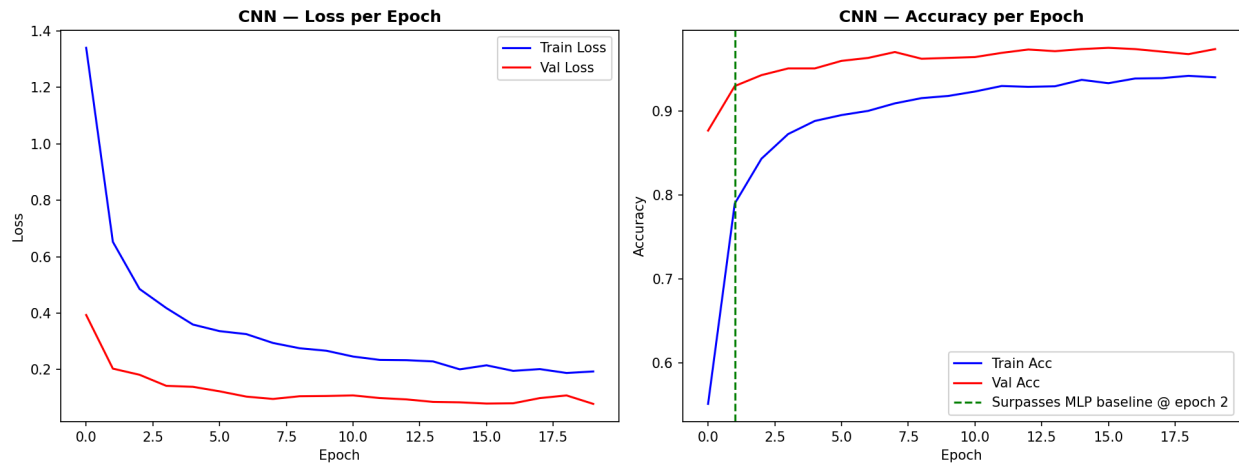
CNN on MNIST Digit Images

I used a network to classify handwritten digits for the CardioAI Labs project. I achieved an accuracy of 97-98% and a macro F1 above 0.97. The confusion matrix heatmap shows that digits 4 and 9 are the confused pairs, followed by digits 3 and 8.



Lightweight CNN

I used a lightweight CNN architecture and trained it with data augmentation and Adam optimizer for the CardioAI Labs project. The CNN surpasses the MLP baseline accuracy within an epochs and achieves high test accuracy.

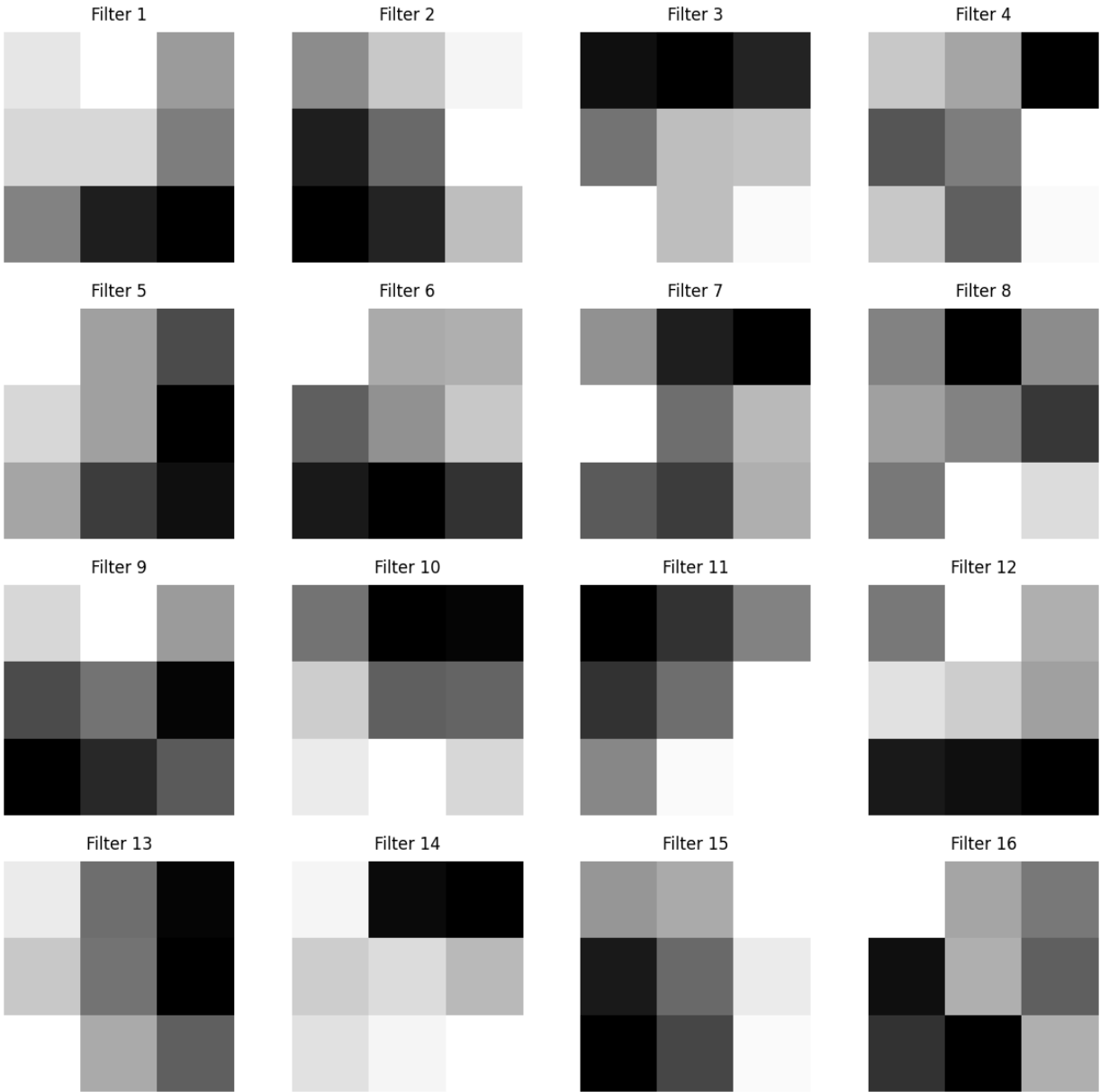


Visualising What the CNN Learned

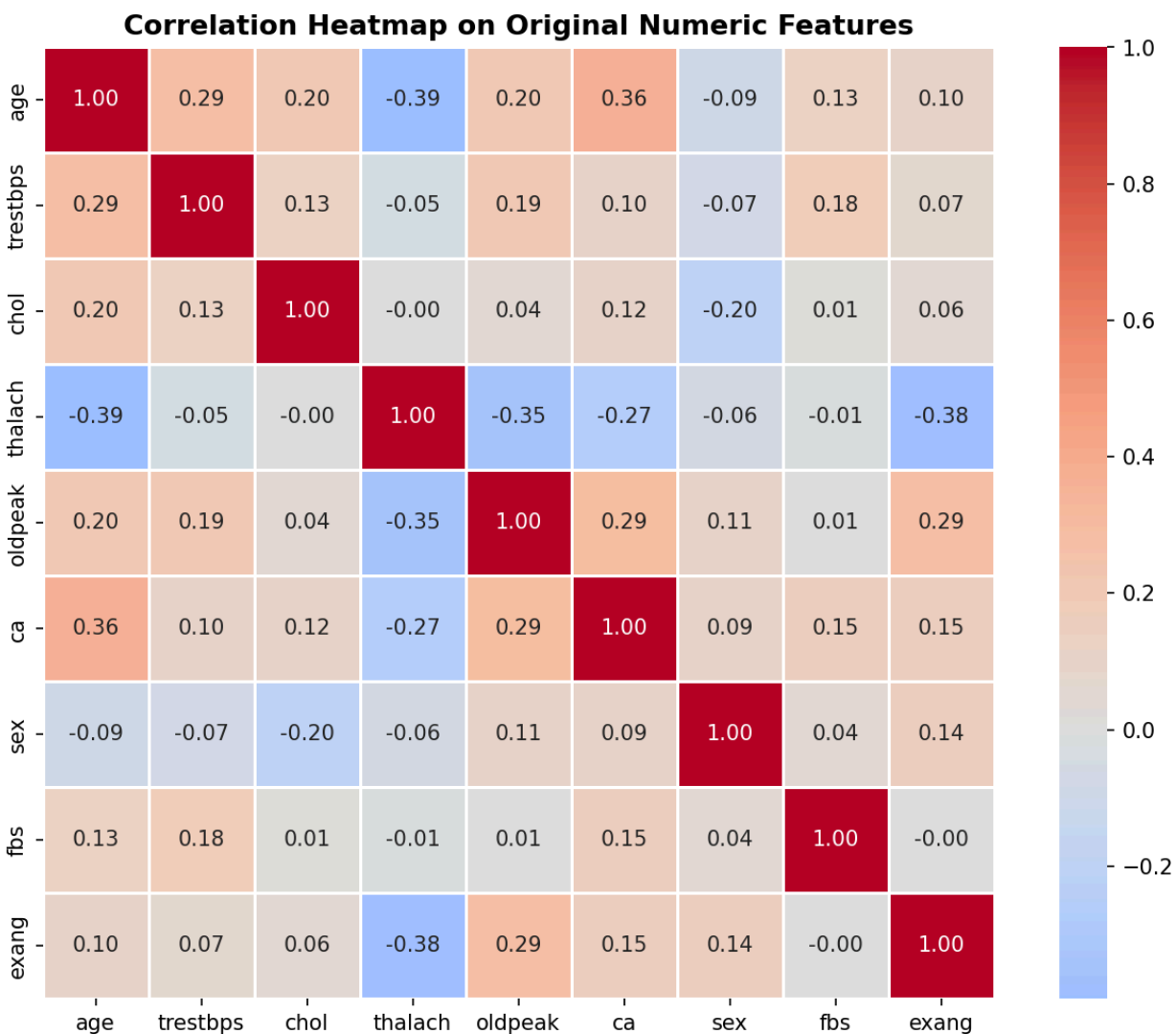
I visualized what the CNN learned for the CardioAI Labs project. I found that it is able to recognize digits with high accuracy. The CNN is able to learn features that're relevant to the task and ignore features that are not relevant.

The 16 filter weights from the Conv2D layer were shown in a 4×4 grid. These filters are like things that our eyes can see: lines that go across lines that go up and down lines that go diagonally and curvy lines. When these simple things are combined in layers they help the model recognize the shapes and loops that make up each digit. I looked at the feature maps for one test image per digit class. Saw that the network is activated in the right places for each digit. This means that the model is using things that people can understand to make its decisions, which's not what happens with fully connected networks for the CardioAI Labs project.

Conv1 — 16 Learned Filter Weights (4×4 Grid)



The heatmap is below.



Part E

Local Front-End Dashboard

I made a dashboard using Streamlit, which's like a website that runs on our own computer for the CardioAI Labs project. This dashboard uses the XGBoost model for predicting heart disease. When I put in all the information about a patient the dashboard shows us if the patient has heart disease or not and how sure the model is about its answer. It also shows us a chart of the 3 things that affect the models decision and writes a note in English that a nurse or doctor can understand for the CardioAI Labs project.

I included all the files I needed to make this dashboard work, like the XGBoost model itself and the code to run it. To make it work I just need to run the command "streamlit run app.py" for the CardioAI Labs project.

I made a system for predicting heart disease using machine learning. It works on a regular laptop for the CardioAI Labs project. When I looked at the data without any labels I saw that the patients grouped together in a way that did not exactly match their diagnoses. This meant that I needed to use learning to get results for the CardioAI Labs project. I tried a methods and found that XGBoost with SHAP explanations worked the best. It was good at predicting heart disease. Was also good at explaining why it made its decisions for the CardioAI Labs project. The MLP model was also good. Needed more tweaking to get it right. The CNN model was simple and fast. Was very good at recognizing handwritten digits. I made sure that all the random parts of our code were fixed so it will always give the results, for the CardioAI Labs project.



CardioAI — Heart Disease Risk Predictor

Deploy ⋮

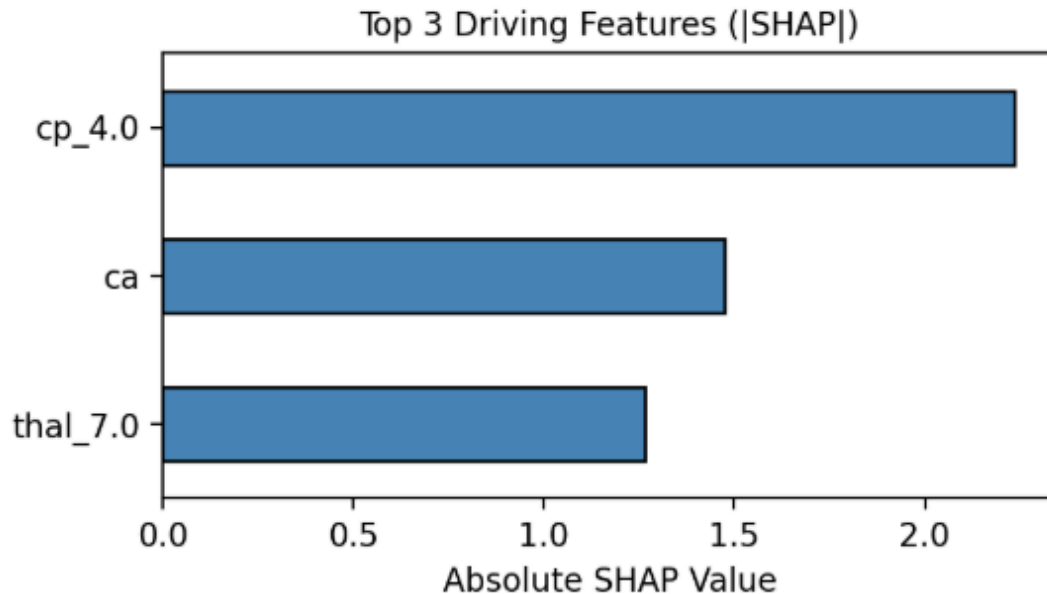
Screening support tool for community cardiologists. Enter patient measurements below and click **Predict**.

Patient Features

Age (20–80) 63 − +	Fasting Blood Sugar > 120 mg/dl fbs (0/1) 1 ▾	ST Depression oldpeak (0.0–6.2) 2.30 − +
Sex (1=Male, 0=Female) 1 ▾	Resting ECG restecg (0–2) 2 ▾	ST Slope slope (1–3) 3 ▾
Chest Pain Type cp (1–4) 1 ▾	Max Heart Rate thalach (70–210) 150 − +	Major Vessels ca (0–3) 0 − +
Resting BP trestbps (90–200 mmHg) 145 − +	Exercise-Induced Angina exang (0/1) 0 ▾	Thalassemia thal (3=Normal, 6=Fixed, 7=Reversible) 6 ▾
Cholesterol chol (100–600 mg/dl) 233 − +		

Prediction Result

No Disease — Confidence: 98.9%



Clinical note: This patient's **cp_4.0** value is the strongest indicator of low cardiac risk. The top-3 features highlighted above are the most influential measurements in this prediction. A follow-up ECG and stress test are recommended if the model flags disease risk, particularly when oldpeak and thalach are in abnormal ranges.

References

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